COMP 7500 Advanced Operating Systems
Project 3: AUbatch - A Pthread-based Batch Scheduling System

Portable Compilation of C Code

Below are some general notes on portable compilation of C code. The tux lab machines may be used to validate against all of the following portability issues.

For loop initial declaration

For loop initial declaration is a C99 feature that is not present in earlier versions of C. Opposed to writing loops like:

```
int i;

for (i = 0; i < 10; i++) { ... }
```

It allows you to write them as

```
for (int i = 0; i < 10; i++) { ... }
```

Different compilers use different auto-detection / defaults for interpreting syntax. If you use this feature you need to specify the version of the C programming language you are using to the compiler with the `-std=c99` flag. Otherwise, some compilers will not understand this syntax implicitly and raise errors. Specifying the flag ensures all compilers will interpret the syntax in the same way.

Linking with external libraries

Libraries like `pthread` and some others need to be linked during compilation.

- To link with `pthread`, you need the `-lpthread` flag in your compilation commands.
- If you use functions like `pow`, then you need to link with the math library using the `-lm` flag.

POSIX and BSD functions

You need to explicitly declare when you are using POSIX / BSD / GNU extensions. There are different ways to do this, but the simplest is to insert following definition above the include for `<stdio.h>`.

POSIX

`getline` is an example of a POSIX extension and requires:

```
#define  _POSIX_C_SOURCE 200809L

#include <stdio.h>
```

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BSD

`usleep` is an example of a BSD extension and requires the following. It is equivalent to the above, but with some added BSD functions.

```
#define _BSD_SOURCE

#include <stdio.h>
```

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GNU

Please refer to the man pages for `getline`, which actually uses a different declaration of `_GNU_SOURCE` that is typically frowned upon for not being as portable as using either POSIX or BSD extensions. `_GNU_SOURCE` combines both of the above declarations with additional GNU extensions that may be unstable, or otherwise duplicate existing functionality reference.